

Week 5 Report: Advanced Models, Tuning & Comparison

Ayomide Bello

Overview

This week moved from a single baseline classifier to comparing three algorithms — Decision Tree, Random Forest, and Gradient Boosting — on the same Titanic survival dataset used since Week 1, then tuning the strongest performer with GridSearchCV and 5-fold cross-validation.

Dataset and Preparation

891 Titanic passengers, 8 features after cleaning (Pclass, Age, SibSp, Parch, Fare, Sex, Embarked). Cabin was dropped (over 70% missing); Age and Embarked gaps were filled with median and mode respectively. Sex and Embarked were one-hot encoded. Data was split 80/20 (712 train / 179 test), stratified to preserve the survival ratio in both sets.

Baseline Model Comparison

All three models were trained with default parameters and evaluated identically.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Decision Tree	0.821	0.794	0.725	0.758	0.795
Random Forest	0.816	0.781	0.725	0.752	0.834
Gradient Boosting	0.799	0.789	0.652	0.714	0.818

No single model won on every metric. Decision Tree had the highest raw accuracy, but Random Forest had the highest ROC-AUC — a better measure of how well a model separates the two classes across all thresholds, not just the default 0.5 cutoff. Gradient Boosting trailed both at default settings. Random Forest was selected for tuning as the more robust overall performer.

Hyperparameter Tuning

Random Forest was tuned using GridSearchCV with 5-fold cross-validation, optimizing for ROC-AUC, across 108 parameter combinations:

- n_estimators: 50, 100, 200
- max_depth: 3, 5, 10, None
- min_samples_split: 2, 5, 10
- min_samples_leaf: 1, 2, 4

Best parameters found: n_estimators = 100, max_depth = None, min_samples_split = 10, min_samples_leaf = 4.
Best cross-validation ROC-AUC: 0.879.

5-fold cross-validation accuracy on the training set: 0.820 (± 0.025) — a more reliable performance estimate than a single train/test split, since it isn't sensitive to how the data happened to be divided.

Before vs. After Tuning

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Random Forest (default)	0.816	0.781	0.725	0.752	0.834
Random Forest (tuned)	0.810	0.807	0.667	0.730	0.833

Tuning lifted precision ($0.781 \rightarrow 0.807$) and cross-validation ROC-AUC ($0.834 \rightarrow 0.879$), but test-set ROC-AUC came in essentially flat and accuracy dipped slightly. This is a realistic result, not an error: GridSearchCV optimizes against cross-validation folds on training data, and a 179-row test set means individual metrics can shift by a point or two just from which passengers landed in that split. The honest read is that the model was already close to its ceiling on default settings — tuning refined it modestly rather than transforming it.

Key Findings and Recommendations

- Ensemble methods don't automatically beat a single Decision Tree — on this dataset size (712 training rows), results shifted meaningfully between algorithms, which is why checking more than one metric matters.
- Sex and Passenger Class remain the dominant survival predictors across all models, consistent with Week 4's Logistic Regression findings — different algorithms agreeing on what matters is a useful cross-check.
- Tuning produced a modest, not dramatic, improvement — future effort is likely better spent on feature engineering than further hyperparameter search on this dataset.
- Recommendation: the tuned Random Forest is the reasonable choice given its balance of precision, ROC-AUC, and interpretability through feature importances.